Hacker #1: Diamond vs BankHeist

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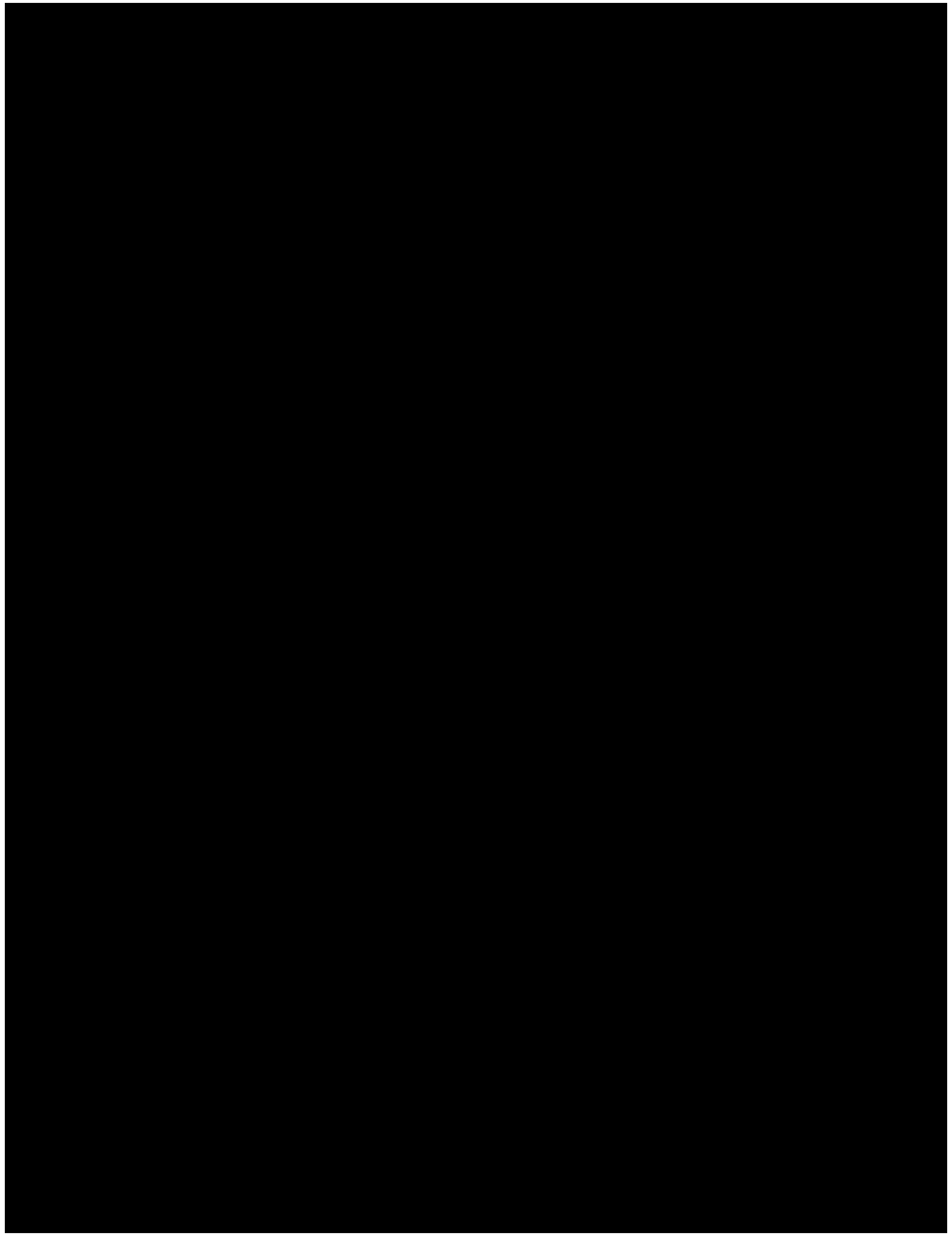
Failure of Diamond

Game	Random	Human	SimPLe	TWM	IRIS	DreamerV3	STORM	DIAMOND (ours)
Alien	227.8	7127.7	616.9	674.6	420.0	959.0	**983.6**	744.1
Amidar	5.8	1719.5	74.3	121.8	143.0	139.0	204.8	**225.8**
Assault	222.4	742.0	527.2	682.6	1524.4	706.0	801.0	**1526.4**
Asterix	210.0	8503.3	1128.3	1116.6	853.6	932.0	1028.0	**3698.5**
BankHeist	14.2	753.1	34.2	466.7	53.1	**649.0**	641.2	19.7
BattleZone	2360.0	37187.5	4031.2	5068.0	13074.0	12250.0	**13540.0**	4702.0
Boxing	0.1	12.1	7.8	77.5	70.1	78.0	79.7	**86.9**
Breakout	1.7	30.5	16.4	20.0	83.7	31.0	15.9	**132.5**
ChopperCommand	811.0	7387.8	979.4	1697.4	1565.0	420.0	**1888.0**	1369.8
CrazyClimber	10780.5	35829.4	62583.6	71820.4	59324.2	97190.0	66776.0	**99167.8**
DemonAttack	152.1	1971.0	208.1	350.2	**2034.4**	303.0	164.6	288.1
Freeway	0.0	29.6	16.7	24.3	31.1	0.0	**33.5**	33.3
Frostbite	65.2	4334.7	236.9	**1475.6**	259.1	909.0	1316.0	274.1
Gopher	257.6	2412.5	596.8	1674.8	2236.1	3730.0	**8239.6**	5897.9
Hero	1027.0	30826.4	2656.6	7254.0	7037.4	**11161.0**	11044.3	5621.8
Jamesbond	29.0	302.8	100.5	362.4	462.7	445.0	**509.0**	427.4
Kangaroo	52.0	3035.0	51.2	1240.0	838.2	4098.0	4208.0	**5382.2**
Krull	1598.0	2665.5	2204.8	6349.2	6616.4	7782.0	8412.6	**8610.1**
KungFuMaster	258.5	22736.3	14862.5	24554.6	21759.8	21420.0	**26182.0**	18713.6
MsPacman	307.3	6951.6	1480.0	1588.4	999.1	1327.0	**2673.5**	1958.2
Pong	-20.7	14.6	12.8	18.8	14.6	18.0	11.3	**20.4**
PrivateEye	24.9	69571.3	35.0	86.6	100.0	882.0	**7781.0**	114.3
Qbert	163.9	13455.0	1288.8	3330.8	745.7	3405.0	**4522.5**	4499.3
RoadRunner	11.5	7845.0	5640.6	9109.0	9614.6	15565.0	17564.0	**20673.2**
Seaquest	68.4	42054.7	683.3	**774.4**	661.3	618.0	525.2	551.2
UpNDown	533.4	11693.2	3350.3	**15981.7**	3546.2	9234.0	7985.0	3856.3
#Superhuman (↑)	0	N/A	1	8	10	9	10	**11**
Mean (↑)	0.000	1.000	0.332	0.956	1.046	1.097	1.266	**1.459**
IQM (↑)	0.000	1.000	0.130	0.459	0.501	0.497	**0.636**	**0.641**

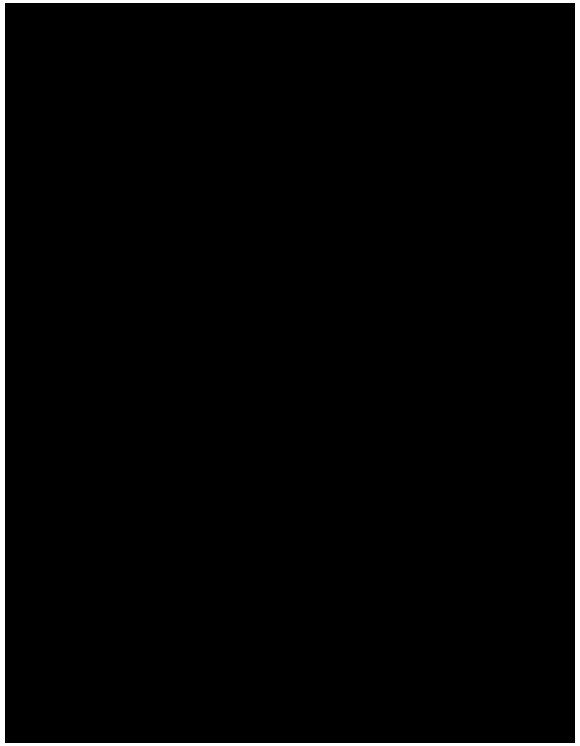
Failure cases

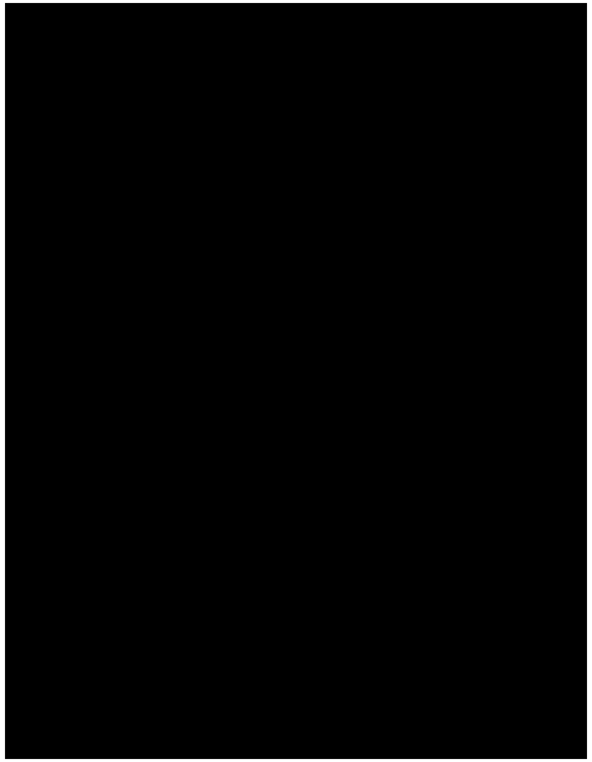
- Policy mode in world model
  - Static scene
- Human mode in world model
  - Weak action response
  - Car disappearing or multiple cars exist simultaneously



Approach 0

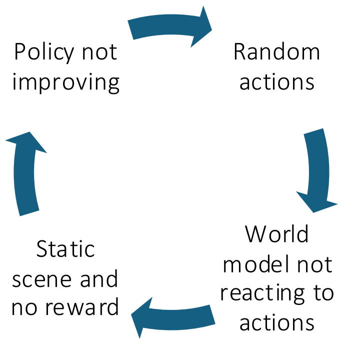
- Finetune DIAMOND on BankHeist for 20k steps
  - Avg. return for 25 episodes: 13.2
  - Avg. return for 25 episodes with pretrained Diamond: 12





Further investigation

- Action patterns in BankHeist
  - Human playing: sparse actions
  - Policy playing: random and fast-changing actions
    - Not observing physical limits
    - Following command intercepts earlier ones



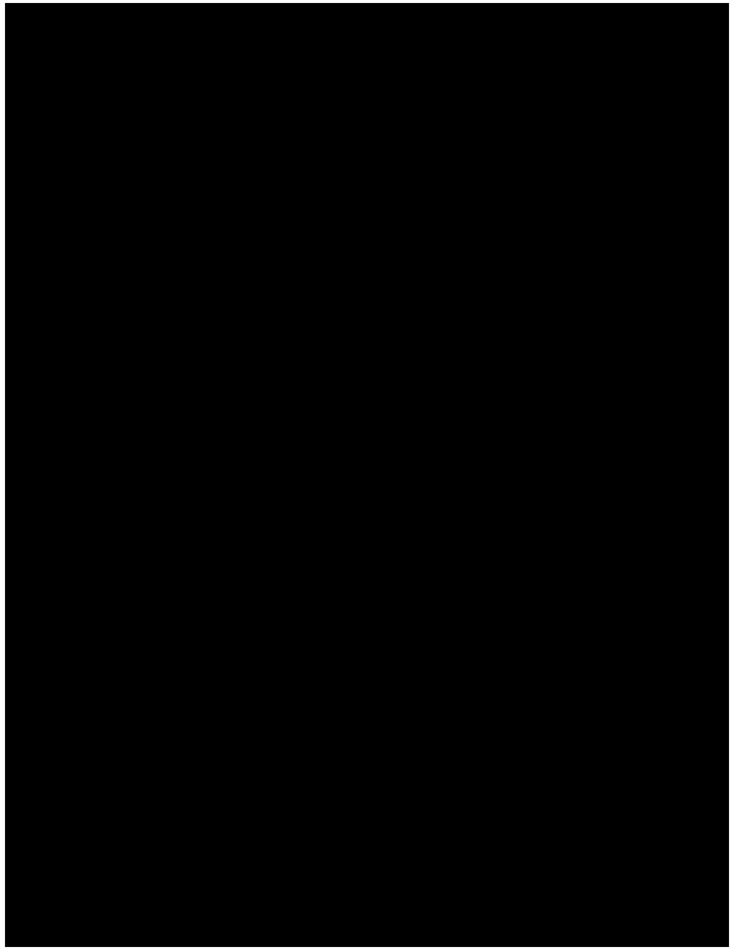
*The car will move along the direction until blocked by a wall or new command received

305	-0.6983	0.4493	0	0.00	0	0
306	-0.6986	0.4493	0	0.00	0	0
307	-0.6985	0.4492	2	0.00	0	0
308	-0.6984	0.4493	2	0.00	0	0
309	-0.6983	0.4494	0	0.00	0	0
310	-0.6983	0.4493	0	0.00	0	0
311	-0.6981	0.4495	0	0.00	0	0
312	-0.6984	0.4493	0	0.00	0	0
313	-0.6983	0.4493	0	0.00	0	0
314	-0.6982	0.4495	0	0.00	0	0
315	-0.6983	0.4494	0	0.00	0	0
316	-0.6986	0.4492	0	0.00	0	0
317	-0.6985	0.4493	0	0.00	0	0
318	-0.6984	0.4493	0	0.00	0	0
319	-0.6983	0.4494	3	0.00	0	0
320	-0.6980	0.4495	3	0.00	0	0
321	-0.6979	0.4496	3	0.00	0	0
322	-0.6978	0.4496	0	0.00	0	0
323	-0.6977	0.4499	0	0.00	0	0
324	-0.6976	0.4498	0	0.00	0	0
325	-0.6974	0.4498	0	0.00	0	0
326	-0.6978	0.4495	0	0.00	0	0
327	-0.6977	0.4495	0	0.00	0	0
328	-0.6976	0.4497	0	0.00	0	0
329	-0.6974	0.4498	0	0.00	0	0
330	-0.6978	0.4496	0	0.00	0	0
331	-0.6977	0.4498	0	0.00	0	0
332	-0.6976	0.4499	0	0.00	0	0
333	-0.6977	0.4498	0	0.00	0	0
334	-0.6980	0.4497	0	0.00	0	0
335	-0.6979	0.4497	0	0.00	0	0
336	-0.6978	0.4498	0	0.00	0	0
337	-0.6977	0.4498	0	0.00	0	0
338	-0.6980	0.4497	0	0.00	0	0
339	-0.6979	0.4497	0	0.00	0	0
340	-0.6978	0.4497	0	0.00	0	0
341	-0.6977	0.4497	0	0.00	0	0
342	-0.6978	0.4495	0	0.00	0	0
343	-0.6974	0.4498	5	0.00	0	0
344	-0.6981	0.4495	5	$\boxed{20.00}$	0	0
345	-0.6985	0.4492	0	0.00	0	0
346	-0.6987	0.4493	0	0.00	0	0
347	-0.6984	0.4495	0	0.00	0	0
348	-0.6988	0.4492	0	0.00	0	0
349	-0.6985	0.4493	0	0.00	0	0
350	-0.6985	0.4491	0	0.00	0	0
351	-0.6981	0.4493	0	0.00	0	0
352	-0.6984	0.4491	0	0.00	0	0
353	-0.6983	0.4491	0	0.00	0	0
354	-0.6984	0.4492	0	0.00	0	0
355	-0.6981	0.4492	0	0.00	0	0
356	-0.6984	0.4491	0	0.00	0	0
357	-0.6986	0.4490	0	0.00	0	0
358	-0.6986	0.4489	0	0.00	0	0
359	-0.6984	0.4492	3	0.00	0	0
360	-0.6983	0.4492	3	0.00	0	0
361	-0.6978	0.4495	0	0.00	0	0
362	-0.6979	0.4494	0	0.00	0	0
363	-0.6980	0.4493	5	0.00	0	0
364	-0.6987	0.4490	5	$\boxed{30.00}$	0	0
365	-0.6988	0.4491	0	0.00	0	0
366	-0.6992	0.4488	0	0.00	0	0
367	-0.6995	0.4488	0	0.00	0	0
368	-0.6995	0.4488	0	0.00	0	0
369	-0.6993	0.4489	0	0.00	0	0
370	-0.6995	0.4489	0	0.00	0	0
371	-0.6997	0.4487	0	0.00	0	0
372	-0.6997	0.4487	[illegible]	0.00	0	0

304	-0.3098	0.4029	3	0.00	0	0
305	-0.3098	0.4029	5	0.00	0	0
306	-0.3098	0.4029	4	0.00	0	0
307	-0.3098	0.4029	16	0.00	0	0
308	-0.3098	0.4029	17	0.00	0	0
309	-0.3097	0.4030	17	0.00	0	0
310	-0.3097	0.4030	10	0.00	0	0
311	-0.3097	0.4030	12	0.00	0	0
312	-0.3097	0.4030	16	0.00	0	0
313	-0.3097	0.4030	0	0.00	0	0
314	-0.1999	0.4631	11	0.00	0	0
315	-0.3097	0.4030	8	0.00	0	0
316	-0.4409	0.3629	10	0.00	0	0
317	-0.7040	0.4166	14	0.00	0	0
318	-0.1998	0.4631	5	0.00	0	0
319	-0.1999	0.4631	12	0.00	0	0
320	-0.2000	0.4631	13	0.00	0	0
321	-0.2000	0.4631	16	0.00	0	0
322	-0.2000	0.4631	15	0.00	0	0
323	-0.2000	0.4631	4	0.00	0	0
324	-0.1999	0.4631	14	0.00	0	0
325	-0.3097	0.4030	9	0.00	0	0
326	-0.4410	0.3629	16	0.00	0	0
327	-0.7041	0.4165	8	0.00	0	0
328	-0.1999	0.4631	14	0.00	0	0
329	-0.3096	0.4029	4	0.00	0	0
330	-0.3096	0.4029	7	0.00	0	0
331	-0.3096	0.4029	14	0.00	0	0
332	-0.3096	0.4029	4	0.00	0	0
333	-0.3096	0.4029	2	0.00	0	0
334	-0.1997	0.4630	9	0.00	0	0
335	-0.3096	0.4029	3	0.00	0	0
336	-0.4409	0.3628	15	0.00	0	0
337	-0.7042	0.4165	11	0.00	0	0
338	-0.1997	0.4630	9	0.00	0	0
339	-0.1997	0.4630	0	0.00	0	0
340	-0.1997	0.4630	16	0.00	0	0
341	-0.1997	0.4630	3	0.00	0	0
342	-0.1997	0.4630	15	0.00	0	0
343	-0.1997	0.4630	13	0.00	0	0
344	-0.1997	0.4630	10	0.00	0	0
345	-0.1996	0.4630	5	0.00	0	0
346	-0.3096	0.4029	4	0.00	0	0
347	-0.4409	0.3628	3	0.00	0	0
348	-0.7040	0.4168	12	0.00	0	0
349	-0.1997	0.4630	8	0.00	0	0
350	-0.1998	0.4630	14	0.00	0	0
351	-0.1998	0.4630	7	0.00	0	0
352	-0.1998	0.4630	10	0.00	0	0
353	-0.1998	0.4630	4	0.00	0	0
354	-0.1998	0.4630	6	0.00	0	0
355	-0.1997	0.4630	2	0.00	0	0
356	-0.3096	0.4028	16	0.00	0	0
357	-0.4409	0.3628	3	0.00	0	0
358	-0.7039	0.4167	12	0.00	0	0
359	-0.1996	0.4629	14	0.00	0	0
360	-0.1995	0.4629	11	0.00	0	0
361	-0.1995	0.4630	15	0.00	0	0
362	-0.1995	0.4629	16	0.00	0	0
363	-0.1995	0.4629	3	0.00	0	0
364	-0.1994	0.4630	14	0.00	0	0
365	-0.3094	0.4029	6	0.00	0	0
366	-0.4407	0.3629	1	0.00	0	0
367	-0.7040	0.4167	13	0.00	0	0
368	-0.1995	0.4630	12	0.00	0	0
369	-0.1995	0.4630	15	0.00	0	0
370	-0.1995	0.4629	0	0.00	0	0
371	-0.1994	0.4630	13	0.00	0	0
372	-0.1995	0.4630	17	0.00	0	0

Approach 1

- Insert *noops (i.e. repeated actions)* between actions
  - --action-repeat 8
  - Avg. return for 25 episodes: 20
    - 19.7 in the paper
    - 12 for 25 episodes
  - Visually more stable actions

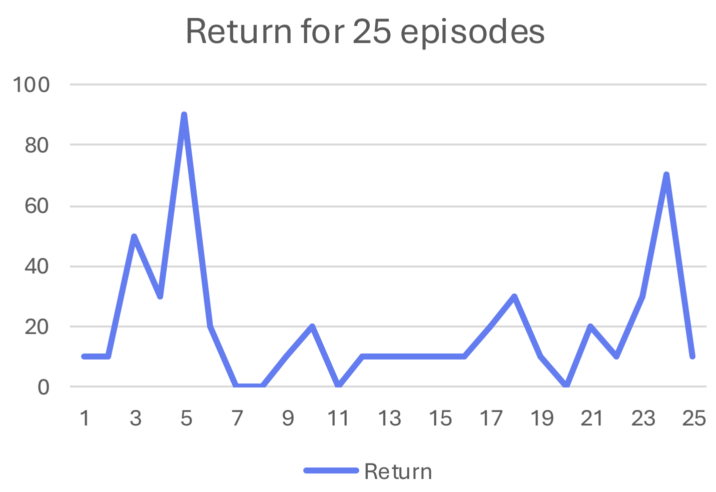


Approach 2

- Add a latent state encoder
  - Frame encoder – reuses Diamond's conv encoder for each frame
  - Action embedding
  - GRU
- Condition the diffusion on the latent state instead of the stacked frames and history actions
- Increase history length from 4 to 8
- Biased sampling – more samples from high reward episodes
- Trained from scratch for 10k steps
- --action-repeat 8

Approach 2 Results

- Avg. return 19.6
  - On par with the original Diamond 19.7
  - Avg. return with pretrained Diamond for 25 episodes 12
- Average #steps per episode increases
  - 1121 vs 614



Future Directions & Lessons Learned

- Integrate the repetitive action constraints to the training process
- Diamond is good at capturing visual details, but not lower-level abstract dynamics/states
  - e.g. Physical constraints, action patterns
  - Can we better complement diffusion world model by capturing and supplying abstract world states?
- Thought exercise: targeted performance vs generalizability